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On Building Real Time Intelligent Agricultural Commodity Trading Models

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***Abstract*— The global population is expected to reach 9.7 billion in 2050, according to the UN News [1]. This suggests that food, agriculture, and global commodity trading are essential and critical to human beings. To stabilize food price and better understand and predict the market trend, people need a real-time efficient online platform as an intelligent trading tool to support dynamic agriculture commodity trading. However, current trading platforms lack AI-powered solutions to support traders and investors with dynamic commodity market analysis based on historical big data, accurate forecasts on market trends and trading price forecasts. This paper focuses on this demand by studying and comparing existing machine learning models for intelligent trading and price prediction. The paper presents three improved machine learning models based on LSTM, GRU, and SARIMA, and proposes an advanced deep learning LSTM model with time series and multi-variant features to forecast crop market trends with predicted prices. As a result, our proposed agriculture trading models for crops (corn, oats, soybean, and soybean oil) based the improved LSTM provide great accurate prediction results (>0.95%). Based on our comparative evaluation results, the proposed model outperforms other existing models. The research results show a great potential application for a future intelligent trading platform which provides traders with real-time data driven market analysis and accurate predictions on market trends and commodity trading prices.**

Keywords—smart agriculture, intelligent agriculture commodity trading, commodity trading price prediction

I. Introduction

The agriculture commodity market is used as an economic leverage by the governments for grain reserves, spot trading guidance, risk avoidance and market stabilization. FIA[2] reports that more than 2.5 billion worldwide contracts of agriculture futures and options were traded in 2020. According to market surveys [3], the agricultural commodity futures market will continue to expand in the next three decades, and the market value will reach 4 billion by 2026. In an online commodity trading market, a quick response and correct trading decision usually is the key to making a successful trading and great profits. However, the current online commodity trading platforms lack of real-time crop production and analysis information, and historical trading price analysis and price prediction tools, therefore, traders usually passively receive information about the futures or options they hold, and then make their own trading decisions

There are numerous research papers focus on commodity trading market analysis and price prediction based on historical price using statistics approaches. Most of the current research work used historical price as the single input feature to predict price. However, the commodity market is influenced by various factors, including weather political decisions and events, economic environment, technology advances, commodity production, as well as trading price. Hence, considering historical price as the single factor is not good enough to accurately predict the commodity trading price due to the market volatility.

This paper attempt to consider multiple input factors in commodity price prediction using data-driven machine learning models and approaches. To select the closely related factors (or features), we use a heatmap to analyze the correlation coefficient of related features and select the closely related variables as the inputs of each crop for price prediction. Our study area covered four different crops (corn, oats, soybeans, and soybean oil). This paper represents our comparative research results between single-input feature price prediction and multiple -input feature price prediction using both statistic models and deep learning models.

In this paper, we present three improved machine learning models based on LSTM, GRU, and RNN, and propose an advanced deep learning LSTM model with time series and multi-variant features to forecast crop market trends with predicted prices. As a result, our proposed agriculture trading models for crops based the improved LSTM provide great accurate prediction results (>0.95%). Based on our comparative evaluation results, the proposed model outperforms other existing models. The research results show a great potential application for a future intelligent trading platform which provides traders with real-time data driven market analysis and accurate predictions on market trends and commodity trading prices.

Rest of the paper has been organized and presented as below: Section II, relative work on prediction models and existing agricultural intelligence and trading platforms. Section III, detailed description of data preparation. Section IV, Methodology description of proposed models. Section V, experimental results analysis.

II. Related work

A. Existing Research on Intelligent Trading Models

Crop price prediction is an important feature for commodity trading platforms to support clients to make smart and right decisions in commodity trading in order to deal with dynamic market. Building crop price prediction needs

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accurate real-time prediction models. This section reviews the existing machine learning models in commodity price prediction. In general, there are two types of models in price prediction: a) statistic based models, and b) deep learning models.

In the past, some statistic models have been developed. For example, Kwas, M., & Rubaszek, M.[4] use partial projection regression model to predict wheat commodity prices. They challenge random walk, no-change forecast as the benchmark on forecasting commodity price and get a conclusion that random walk benchmark should be substituted in predict real commodity price. Xiong, T., Li, C.[5] use a combination method based on regression models, such as vector error correction model and multi-output support vector regression (VECM-MSVR) to forecast the interval of agricultural commodity prices. Among these models, ARIMA (Autoregressive Integrated Moving Average) performs well in short-term forecasting, but it's not designed for long-term prediction.

There are numbers of published paper discussed deep learning models for price prediction. For example, Zhiyuan Chen, Howe Seng Goh[6] compared five famous popular models, the Autoregressive Integrated Moving Average (ARIMA), Support Vector Regression (SVR), Prophet, XGBoost and LSTM(Long short-term memory). Based on their results, LSTM is the most reliable mode with an average of 0.304 mean-square error. Ouyang, H., Wei, X., & Wu, Q. [7] proposed a new model to handle both long-term and short-term forecasting called (LSTNet). Since the prices of agricultural products are often controlled by the government, the agricultural product price information often has a longer time lag. LSTNet has better performance in dealing with the combination of short-term and long-term data information compared with other predictive models. Helin Yin and Dong Jin[8] used the Loess (STL) preprocessing method based on LSTM, and developed STL-ATTLSTM (STL-Attention-based LSTM). Because they introduce attention into the model, STL-ATTLSTM shows a good performance in the forecast of agricultural commodity prices. Although there are many mature models used for forecasting agricultural commodity price, we still need more price prediction models with higher accuracy and better performance.

Fig.1 lists the relevant papers on commodity price prediction, and the models based on machine learning models.

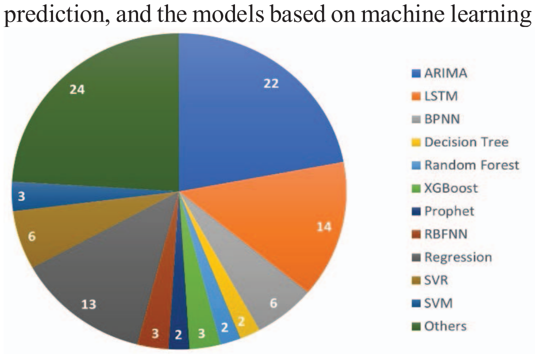

Fig. 1. Model selection by relevant papers

B. *A review of Intelligent platform*

As shown in table 1, six existing agriculture commodity trading platforms are compared, including Aspect, eka, VOSBOR, etc. Current intelligent system platform services focus on commodity trading and decision-making. As shown, all of the six platforms do not provide clients with features in crop production forecasting and commodity price prediction. Aspect Enterprise Solutions[9] and Eka[10] focus on providing users with commodity trading and risk management, commodities data, and decision support. Both platforms support Commodity Trading and Risk Management (CTRM) features. The Climate FieldView™ [11] and Farm Log[12] use big data technology to monitor commodity trading transactions, and providing farmer clients with advices in decision making to maximize their profitability. IBM Watson® Decision Platform[13] combines artificial intelligence (AI) and agricultural Internet of Things (IoT) to provide user with predictive insights based on agricultural IoT data.

TABLE 1 COMPARISON OF EXISTING FIVE EXISTING AGRICULTURAL INTELLIGENCE AND TRADING PLATFORMS

Platform	Feature	Deployment	Target Users
Aspect Enterprise Solutions	Agricultural analytics *CTRM	**Saas Cloud	Commodity Energy trader
Eka	Agricultural analytics *CTRM	**Saas Cloud	Commodity Supplier
Climate FieldView™	Farm Management Artificial Intelligent	**Saas Mobile	Farmer Farm manager
FarmLogs	Farm Management Artificial Intelligent	**Saas Mobile	Farmer
IBM® Watson® Decision Platform	Data Management Artificial Intelligent	**Saas Web Cloud	Farmer Agriculture industry

*CTRM: Commodity Trading and Risk Management
**Saas: software as a service

III. DATA ENGINEERING

The factors impacting crop commodity price are complex combination of dynamic crop predicted production, weather, politics, war, natural resource, and changes of supply and demand. For crop commodity price forecasting, the historical futures prices, predicted production, and data of changes of supply and demand have been collected and to be used as features in calculating Seaborn correlation score to analyze the correlation between features related to production and price to forecast the price trend for corn, soybean, and soybean oil.

A. *Data Source & Pre-Processing*

The Table 2 shows the summary of the data source, data size, and data type.

TABLE 2 DATA SOURCE, DATA TYPE & DATA SIZE

Data	Data Type	Item	Data Size	Data Source
Daily	CSV	Corn	1535 * 7	

Crop Historical Price		Wheat	1542 * 7	Investing.com [14]
		Oats	1512 * 7	
		Soybeans	1536 * 7	
		Soybean Oil	1545 * 7	
		Soybean Meat	1544 * 7	
Monthly Predicted Crop Production	XLS	WASDE Monthly Report	60 Reports	USDA [15]

Historical Commodity Price

We collected the five-year (2016-2021) daily historical futures price of collected crops: corn, wheat, oats, soybeans, soybean oil, and soybean meal from investing.com [14] to train and test models, and the historical price dataset of each selected crop includes date, close price, open price, daily highest price, daily lowest price, volume, and percentage of change.

The dataset of each crop contains some missing values either missing date or missing volume. Dataset contains less than ten missing dates. Therefore, we drop the rows with missing date.

Besides, for the missing values of "Volume" column in each crop's dataset. Fig. 2 shows a part of the missing value of soybean historical price and how it was filled.

For some crop datasets, "Volume" column is missing more than two weeks consecutive dates. Simply dropping the rows with missing value appears a cliff in the graph. Also, using average or mean of volume to fill the missing value does not reflect the fluctuation of volume. Hence, we used the average price to multiply by the changes in percentage for each missing field.

The formular below shows the way of calculating the missing volume on Dec. 2, 2021.

$$\text{Volume} = (105{,}480 + 60{,}010)/2*(1+1.3\%) = 83{,}820.69 \quad (1)$$

27-Dec-21	1,362.50	1,337.50	1,363.50	1,333.75		1.49%	
24-Dec-21	1,342.50	1,342.50	1,342.50	1,342.50		0.79%	
23-Dec-21	1,332.00	1,326.00	1,334.00	1,321.00	60,010.00	0.24%	
22-Dec-21	1,328.75	1,306.00	1,333.25	1,304.75		1.59%	84060.65
21-Dec-21	1,308.00	1,292.00	1,313.50	1,286.50		1.22%	83754.49
20-Dec-21	1,292.25	1,289.50	1,296.25	1,278.50		0.54%	83191.82
17-Dec-21	1,285.25	1,276.25	1,297.50	1,274.75		0.63%	83266.29
16-Dec-21	1,277.25	1,262.00	1,280.75	1,258.75		1.17%	83713.12
15-Dec-21	1,262.50	1,260.00	1,266.50	1,251.00		0.24%	82943.59
14-Dec-21	1,259.50	1,246.00	1,267.00	1,237.50		1.25%	83779.31
13-Dec-21	1,244.00	1,271.00	1,271.00	1,243.25		-1.87%	81197.67
10-Dec-21	1,267.75	1,264.25	1,276.50	1,259.25		0.26%	82960.14
9-Dec-21	1,264.50	1,260.25	1,267.00	1,247.00		0.28%	82976.69
8-Dec-21	1,261.00	1,252.25	1,264.00	1,238.25		0.86%	83456.61
7-Dec-21	1,250.25	1,262.25	1,268.50	1,243.00		-0.89%	82008.57
6-Dec-21	1,261.50	1,271.75	1,274.75	1,254.50		-0.45%	82372.65
3-Dec-21	1,267.25	1,244.50	1,270.00	1,244.25		1.85%	84275.78
2-Dec-21	1,244.25	1,229.75	1,249.00	1,221.00		1.30%	83820.69
1-Dec-21	1,228.25	1,222.00	1,237.00	1,216.25	105,480.00	0.90%	82745
30-Nov-21	1,217.25	1,241.50	1,244.75	1,214.25		-1.95%	

Fig. 2. Missing volume data of Soybean

Crop Production Data Source & Pre-processing

As mentioned above, the predicted crop production and data caused the changes of supply and demand are very important factors impacting the futures price, we also collected five-year crop production related data such as from the World Agricultural Supply and Demand Estimates (WASDE) from United States Department of Agriculture (USDA) [15], which is a monthly report released around the 10^{th} of each month. This report provides very valuable information influencing the changes of the predicted production. Although there are some common closely related features affecting calculating predicted production for the crops, some other features are different based on different crops. We summarized the features that were collected for the four crops as the table 3 below.

TABLE 3: THE SUMMARY OF FEATURES FROM WASDE REPORT

Item	Common Features	Different Features
Oats	Beginning stocks Production Imports Total supply Exports Total consumption Ending stocks	Feed and residual, Total domestic consumption
Corn		Feed and residual, Ethanol & by-products, Total domestic use
Soybeans		Crushing, Seed, Residual
Soybean Oil		Domestic disappearance, Biofuel, Food, feed & other industrial

B. Input Dataset Processing for Machine Learning Models

The final input dataset includes the price related factors and production related factors for each crop. However, since the historical price data is daily based and predicted production data is monthly based, we match the monthly production data with the daily futures price.

We assume every new released monthly WASDE report could cause a new period of price fluctuation, and the impact of it could last for a month until the next monthly report is released. For example, the 2021 November and December reports were released on the 9^{th} of November and December 2021 respectively, we used the corn data from 2021 November report matches the corn daily price data for the period from 11092021-12082021. When 2021 December reports released, the new corn production data we got from the new report was matched with corn daily price data from 12092021 until the date one day before the 2022 January report release date.

Additionally, soybeans, soybean oil, and soybean meal are highly related to each other in supply and demand relationship, so we added the soybean oil price and soybean meal price to the soybeans final input dataset as features. We adopted the same way to complete the soybean oil and soybean meal final input datasets.

Similarly, in the market, as the price of corn rises, the supply of wheat will fall; as the price of corn falls, the supply of wheat will rise. Therefore, we added the wheat price to the corn final input dataset as a feature to calculate the correlation score to filter the features for the machine learning models.

C. Data Parameter Selection for Machine Learning Models

Not all the selected features for the final datasets have the same impact degree to the futures price, we selected top six parameters which have very strong impacts on the futures price either in the positive impact side or negative impact side for each crop by calculating the Seaborn correlation score and use heatmap to visualize it.

The Fig. 3 shows the correlation score of corn as an example. As a result, we selected wheat price, total consumption, consumption ratio, ending stocks, and beginning stocks for corn's price prediction models.

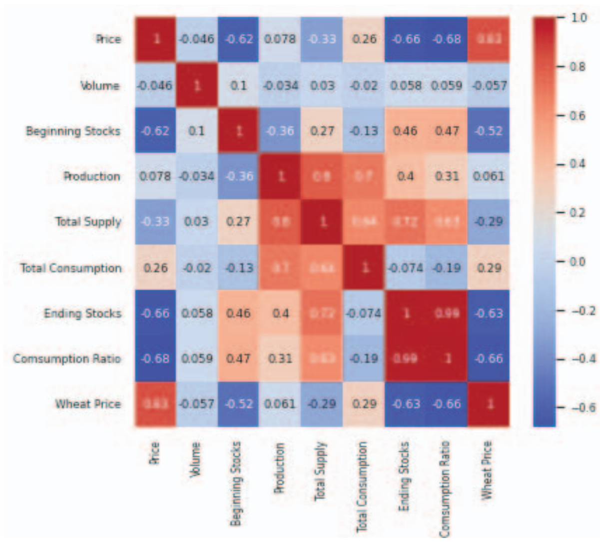

Fig. 3. Corn correlation heatmap

D. Data Segregation

The datasets were partitioned into train, validation, and test sets. As the prediction was based on time series models, the datasets were firstly organized in chronological order. Then the top 60% data were segregate as train set, the next 20% were validation set, and the final 20% were test data.

IV. MACHINE LEARNING MODEL

Two groups of models are used in this research. The first group includes three single-input feature models: simple LSTM, simple GRU, and Prophet. The second group includes four multiple-input feature models, which consider selected input features, such as market inflation, crop production, natural disasters, weather changes, and political events or even wars. In this group, we developed three improved models: a) improved LSTM, b) improved GRU, and c) improved RNN. In addition, a new model, known as Advanced Improved LSTM, is developed as a multi-input feature model to cope with the performance issues in the previous three models. The details about these models are explained below.

A. Simple LSTM

LSTM is a special kind of RNN, capable of learning long-term dependencies. A common LSTM unit is composed of a cell, an input gate, an output gate and a forget gate. The cell remembers values over arbitrary time intervals and the three gates regulate the flow of information into and out of the cell[16]. The equations for the gates in LSTM are Compared with the data stability required by some statistical models, LSTM performs better on unstable time series with more fixed components.

B. Simple GRU

GRU is also a variant of RNN aimed at solving the problem of vanishing gradients. compared to LSTM. It's a leaner variant, which means it offers similar performance while running faster[17]. The main difference between GRU and LSTM is that the GRU bag has two gates, reset and update. GRUs are simpler than LSTMs and are more suitable for smaller datasets. Since our dataset only has over a thousand data for each crop, GRU has been selected and used. And it performs well.

C. Prophet

Prophet is a procedure for forecasting time series data based on an additive model where non-linear trends are fit with yearly, weekly, and daily seasonality, plus holiday effects[18]. It works best with time series that have strong seasonal effects and several seasons of historical data. Prophet is robust to missing data and shifts in the trend, and typically handles outliers well. Due to these reasons, Prophet is selected and applied in this research.

D. Improved LSTM

For the purpose of better reflecting market volatility. We improve the LSTM model by adding multi features as the input of the model to predict the price.

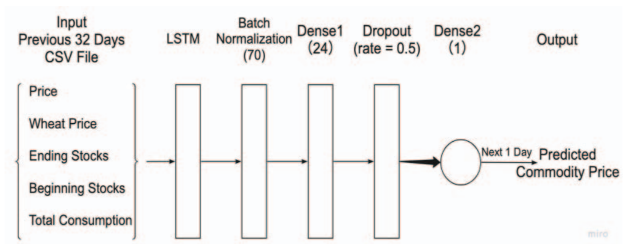

Fig. 4. Architecture of improved LSTM

Fig.4 shows the architecture of our improved LSTM model. The most important improvement is that we provide multi features as input. According to the results displayed by the heat map, after selecting the six features with the highest correlation coefficients, we put the data into model training and testing, and then continuously adjust the model parameters according to the results to obtain the final selection of different features that are most suitable for different crops. Table 4 is the final features we select for each crop.

TABLE 4: FEATURES FOR EACH CROPS

Item	Common Features	Different Features
Oats	Price, Total Consumption, Ending stocks, Beginning Stocks *Price Change (for Advanced Improved LSTM)	Production
Corn		Wheat Price
Soybeans		Soybean Oil price, Soybean meal price
Soybean Oil		Soybean price, Soybean meal price, Production

E. Improved GRU

Since both the GRU model and LSTM are time series deep learning models improved by RNN, we made the updates of GRU by adding multiple-input features based on an correlation analysis. Fig.5 shows the detailed model architecture.

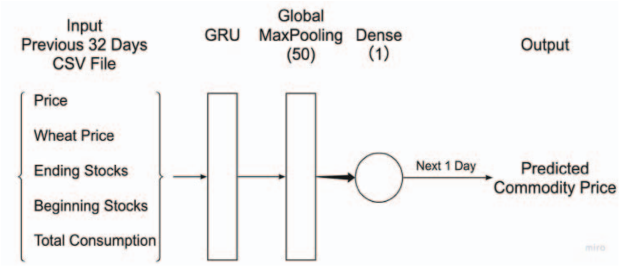

Fig. 5. Architecture of improved GRU

F. Improved RNN

We also set up a RNN model as comparison, the architecture of improved RNN is as Fig. 6.

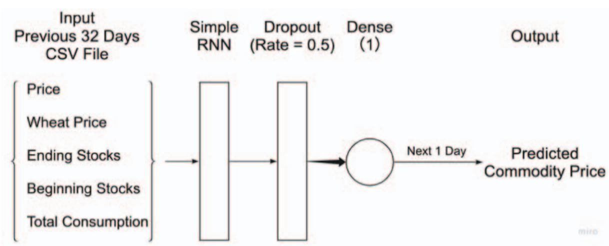

Fig. 6. Architecture of improved RNN

G. Advanced Improved LSTM

Because the performance of the previous three multiple-input feature models does not reach to our expected accuracy, we then develop a new model known as the advanced improved LSTM model based on the existing BI-LSTM model to improve price prediction accuracy. Fig. 7 shows its detailed layered architecture, in which the BI-LSTM model has been used in the first two layers.

The advanced improved LSTM retains the improvements of the improved LSTM, and then optimizes from the aspects of data processing, model architecture and prediction objectives.

First, we found that the price range of agricultural commodity prices varies greatly from year to year due to some global economic reasons. This leads to inability to get good results in price prediction. To solve this problem, we calculate the daily price change and use the price change as an input feature for model training.

When setting up the model, instead of the conventional one-way LSTM, we chose the Bidirectional LSTM model. Figure 4 is the architecture of Bi-LSTM.

As defined, Bbidirectional long-short term memory (BI-LSTM) is the process of making any neural network to have the sequence information in both directions backwards (from future to past) or forward (from past to future) [19]. In a bidirectional model, its inputs can flow in both directions, which means that both past and future information can be preserved.

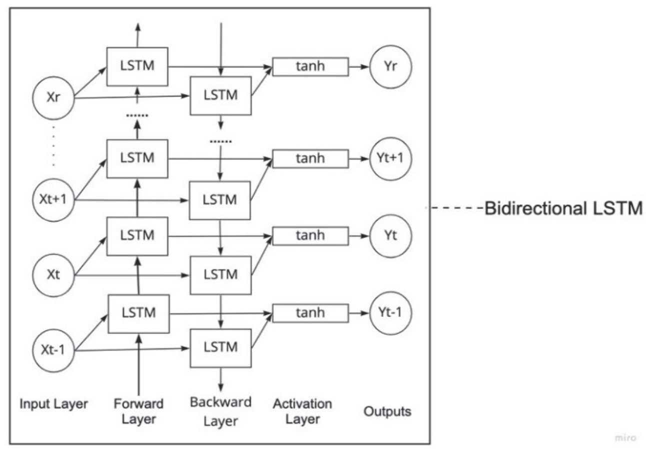

Fig. 7. Bi-LSTM

In this study, we firstly predict the price change for the next day, the next step is adding it to the latest actual price of the day to obtain the predicted price of the next day. Fig. 8 is the architecture of the advanced improved LSTM model. The model has ideal results with great accuracy.

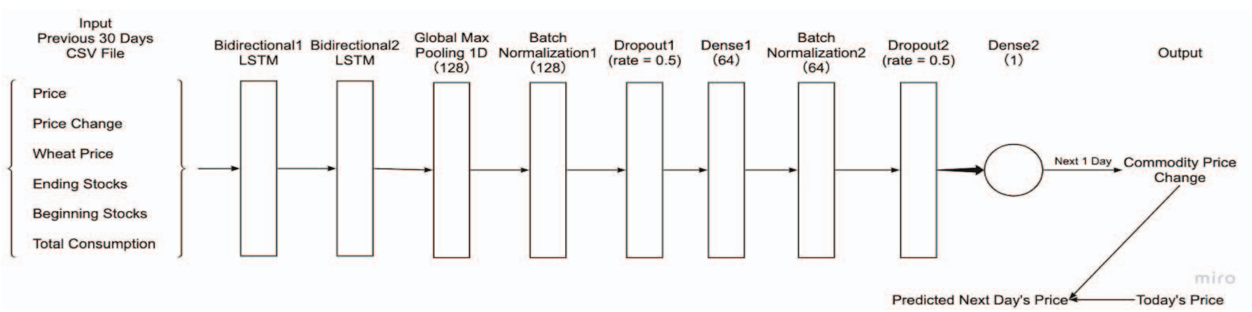

Fig. 8. Architecture of advanced improved LSTM

V. Model Results

Fig. 9 and Fig. 10 show two samples of the results of multi-input models for soybean and oats. We basically used the models to predict the commodity prices for soybean and oats for one year. As shown in Fig. 10, for soybean, the improved GRU model performs a little better than the improved LSTM and the improved RNN. Improved RNN performed the worst, the curve of predicted value if far away from the real value curve. Fig. 9 shows the results of oats, we can see that because of the substantial price growth in late 2021, the prediction didn't give ideal results under LSTM and GRU. RNN performed better for this crop. According to these results, none of improved LSTM, improved GRU and improved RNN performed well for all the four crops; however, the Advanced Improved LSTM model performed the best for all the four crops and the evaluation can also prove that this model is very reliable as a commodity price prediction model.

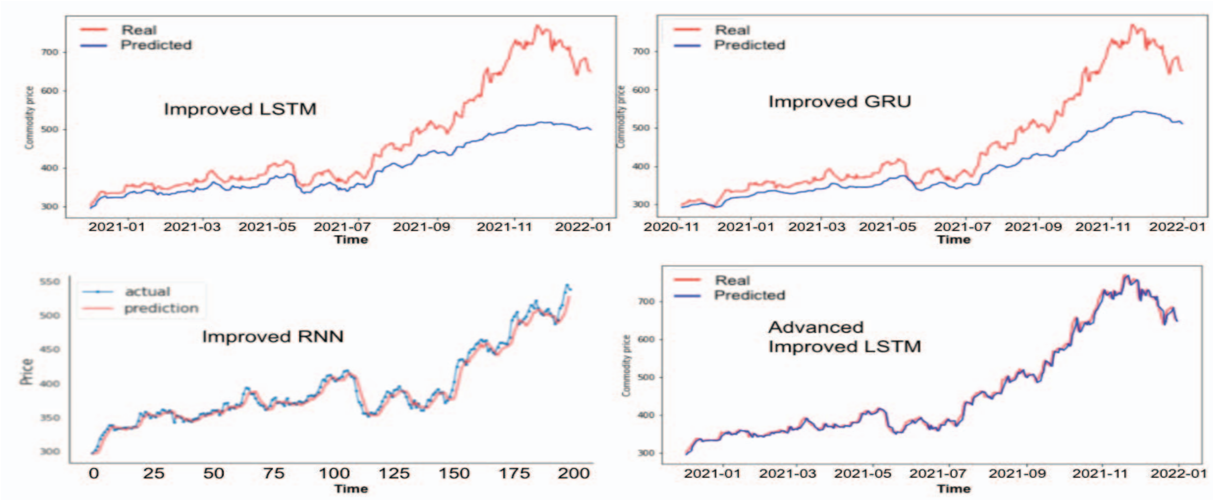

Fig. 9. Results of multi-featuer models-oats

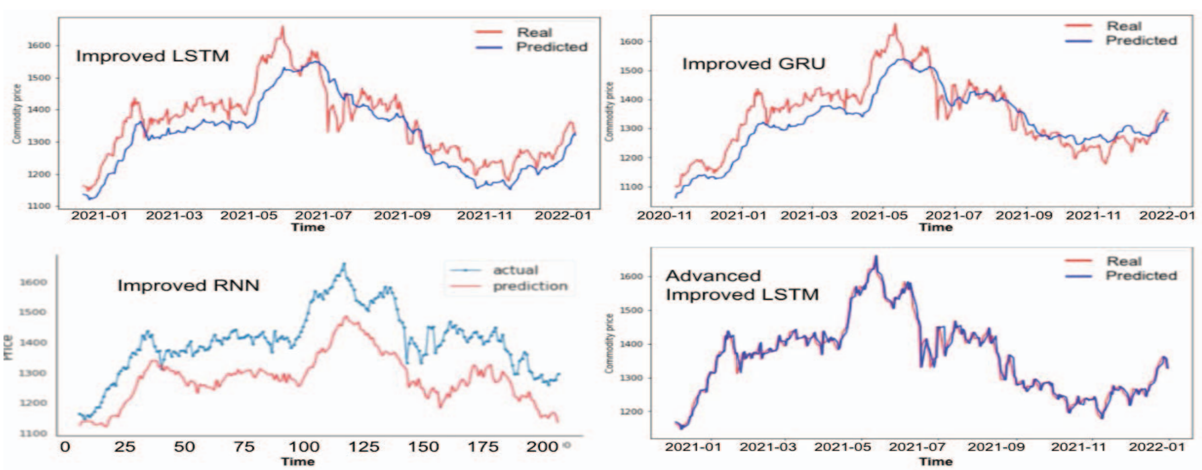

Fig. 10. Results of multi-featuer models-soybeans

Evaluation Method

As a price prediction model, we choose to keep two decimal places in the process from data preprocessing to result prediction. Therefore, the accuracy rate and common classification evaluation methods are not very good criteria for the real performance of our model. Instead, we choose some regression evaluation metrics such as MAE, RMSE and R^2 to evaluate our model.

A. MAE

The mean absolute error (MAE) is the average of the difference between the predicted value and the actual value. The smaller the value of MAE, the closer the predicted value is to the actual value, which means that the model performs better. The calculation is as below:

$$MAE = \frac{\sum_{i=1}^{n} |y_i - x_i|}{n} = \frac{\sum_{i=1}^{n} |e_i|}{n} \tag{2}$$

B. RMSE

The root-mean-square error (RMSE) which is also called the root-mean-square deviation (RMSD) is the square root of the second sample moment of the differences between predicted values and observed values or the quadratic mean of these differences. Similar as MAE, the smaller the RMSE, the better the model performed. The formula is:

$$RMSD = \sqrt{\frac{\sum_{t=1}^{T} (\hat{y} - y_t)^2}{T}} \tag{3}$$

C. R^2

The coefficient of determination, denoted R^2 or r^2, is the square of the coefficient of multiple correlation. Compared with MAE and RMSE, R^2can provide more intuitive information, because the value range of the first two is not fixed, and the value of R^2 is usually between 0 and 1. The closer R^2 is to 1, the better the performance of the model. The formula of R^2 is

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} \tag{4}$$

$$SS_{res} = \sum_i (y_i - f_i)^2 = \sum_i e_i^2 \qquad SS_{tot} = \sum_i (y_i - \bar{y})^2 \tag{5}$$

The following are the evaluation and comparison of single input feature models.

TABLE 5: FEATURES FOR EACH CROPS

Crops	Corn (MAE)	Oats (MAE)	Soybean (MAE)	Soybean Oil (MAE)
LSTM	**0.017**	**0.015**	0.033	**0.018**
GRU	0.025	0.29	**0.022**	0.034
Prophet	16.053	3.927	16.600	1.939

We can tell that LSTM and GRU performs good in single input feature task. The MAE for four crops are all below o.1 under LSTM and GRU model.

Figure 11 and figure 12 are the comparison of multi-input feature models. For multi-input feature models, we chose RMSE and R^2 as evaluation method.

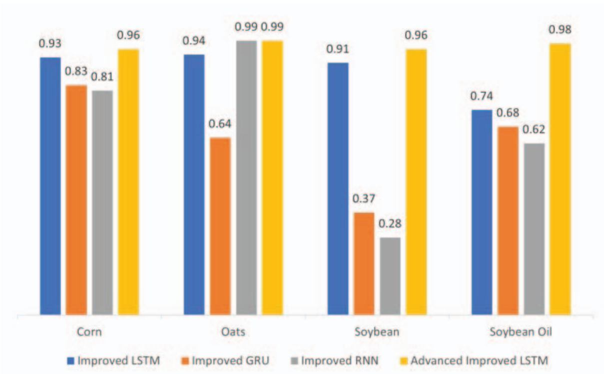

Fig. 11. R^2 comparison

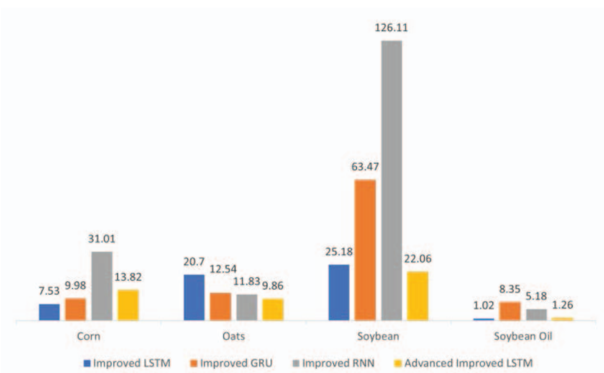

Fig. 12. RMSE comparison

Fig. 11 and Fig. 12 show that the Advanced Improved LSTM has the best overall performance. The R^2 of advanced improved LSTM for four crops are all above 0.95, while the RMSE are all below or near to 20. Compared with Improved LSTM, Advanced Improved LSTM has an average increase of 0.5 in the value of R^2, and an average decrease of 2 in RMSE. This result achieved our expectation and means our prediction model is very reliable.

VI. Conclusion

To predict the accurate price of the agriculture commodity crops, four different crops historical data from 2016 to 2021 are collected, pre-processed, and used to train three sign-input feature models (Simple LSTM, Simple GRU, and Prophet), and four multiple-input feature models (Improved LSTM, Improved RNN, Improved GRU, and Advanced Improved LSTM). Among of them, our proposed AI-LSTM shows the best performance and accuracy by consider multiple input features. In the future, we plan to optimize the advanced improved LSTM model so that it can be used to predict short-term prices, such as 2 to 3 days, or even a week. Furthermore, we plan to consider real-time news in price prediction using text analysis techniques.

References

[1] United Nations. "Growing at a Slower Pace, World Population Is Expected to Reach 9.7 Billion in 2050 and Could Peak at Nearly 11 Billion around 2100 | UN DESA | United Nations Department of Economic and Social Affairs." UN DESA | United Nations Department of Economic and Social Affairs, 18 June 2019, www.un.org/development/desa/en/news/population/world-population-prospects-2019.html. Accessed 27 Sept. 2021.

[2] Global futures and options trading reaches record level in 2020 | FIA. (n.d.). Www.fia.org. Retrieved October 12, 2021, from https://www.fia.org/resources/global-futures-and-options-trading-reaches-record-level-020#:~:text=Washington%2C%20DC%E2%80%94FIA%20today%20released%20yearly%20statistics%20that%20show

[3] Artificial Intelligence in Agriculture Market by Technology (Machine Learning, Computer Vision, and Predictive Analytics), Offering (Software, Hardware, AI-as-a-Service, and Services), Application, and Geography - Global Forecast to 2026. Market Research Firm. (2020, April).

[4] Kwas, M., & Rubaszek, M. (2021). Forecasting commodity prices: Looking for a benchmark. Forecasting, 3(2), 447–459.

[5] Xiong, T., Li, C., Bao, Y., Hu, Z., & Zhang, L. (2015). A combination method for interval forecasting of Agricultural Commodity Futures Prices. Knowledge-Based Systems, 77, 92–102.

[6] Chen, Z., Goh, H. S., Sin, K. L., Lim, K., Chung, N. K. H., & Liew, X. Y. (2021). Automated Agriculture Commodity Price Prediction System with Machine Learning Techniques. Advances in Science, Technology and Engineering Systems Journal, 6(4), 376–384.

[7] Ouyang, H., Wei, X., & Wu, Q. (2019). Agricultural Commodity Futures prices prediction via long- and short-term time series network. Journal of Applied Economics, 22(1), 468–483.

[8] Helin Yin 1,Dong Jin 1,Yeong Hyeon Gu STL-ATTLSTM: Vegetable Price Forecasting Using STL and Attention Mechanism-Based LSTM. (n.d.).

[9] Agricultural Commodity Trading Platform – Aspect Enterprise Solutions. Aspect. (2019, July 11). https://aspectenterprise.com/agriculture/.

[10] Cloud platform for procurement to payments. EKA. (2021, October 8). https://eka1.com/.

[11] Digital Farming's leading software platform. Climate Fieldview. (n.d.). https://climate.com/.

[12] Farm management software and apps. FarmLogs. (n.d.). https://farmlogs.com/.

[13] CIO insights: The Future of Supply Chain Management - ibm.com. (n.d.). https://www.ibm.com/downloads/cas/2BRB2RQM.

[14] Commodities Prices. (n.d.). Investing.com. https://www.investing.com/commodities

[15] the Interagency Commodity Estimates Committees. (n.d.). Publication | World Agricultural Supply and Demand Estimates | ID: 3t945q76s | USDA Economics, Statistics and Market Information System. Usda.library.cornell.edu; The World Agricultural Outlook Board. https://usda.library.cornell.edu/concern/publications/3t945q76s?locale=en#release-items

[16] Thakur, D. (2018, July 6). LSTM and its equations. Medium.

[17] Althelaya, Khaled A., et al. "Stock Market Forecast Using Multivariate Analysis with Bidirectional and Stacked (LSTM, GRU)." IEEE Xplore, 1 Apr. 2018.

[18] Prophet. (2019). Prophet; FaceBook. https://facebook.github.io/prophet/

[19] Verma, Y. (2021, November 20). Complete Guide To Bidirectional LSTM (With Python Codes). Analytics India Magazine. https://analyticsindiamag.com/complete-guide-to-bidirectional-lstm-with-python-codes/